FOR COMPLETE BEGINNERS

One loop, any length.

Complete Modern Java, episode 15: looping over an array, in Java and TypeScript

> Add up every price with one loop, and know the one character that crashes it.

Three prices are three additions. A hundred prices would be a hundred.

One loop adds them all: the loop variable is the index, and it stops before the length.

In this episode

- for with an index
- i < prices.length
- The <= crash
- Enhanced for (for-each)
- An empty array
- for...of in TypeScript
- NaN in TypeScript

Coming next

- Classes and objects (episode 16)

At the time of recording: Java 25, Node 24 LTS, IntelliJ IDEA 2026.1.3.

01 What each run printed

Basket.java: for with i < prices.length	`499`
Basket.java: the same loop with <=	`ArrayIndexOutOfBoundsException: Index 3 out of bounds for length 3`
Basket.java: for (int price : prices)	`499`
Basket.java: int[] prices = {}	`0 and 0`
basket.ts: for and for...of	`499 and 499`
basket.ts: the same loop with <=	`NaN, with no error`

Every row was run, not predicted.

02 The index walks the array

i starts at 0, the first index. The check i < prices.length comes before every pass, so prices[3] is never read.

Change < to <= and i reaches 3: the body reads a box that does not exist, and Java stops.

Basket.java

```
void main() {
    int[] prices = {150, 99, 250};
    int total = 0;
    for (int i = 0; i < prices.length; i++) {
        total += prices[i];
    }
    IO.println(total);                // 499
    int sum = 0;
    for (int price : prices) {
        sum += price;
    }
    IO.println(sum);                  // 499
}
```

> Printed 499 twice; with <= it stopped with Index 3 out of bounds for length 3; with {} it printed 0 twice.
>
> Run on a Mac, 2026-11-06: Java 25.0.4, Node 24.22.0, TypeScript 7.0.2.

03 Which loop?

Want every value? for (int price : prices), read as "for each price in prices": no index, no boundary to get wrong.

Need the position, or to change the array? Use the loop with the index: prices[i] = prices[i] * 2.

Trace of the index loop

i	i < 3	prices[i]	total
			0
0	true	150	150
1	true	99	249
2	true	250	499
3	false	not read	499

> Twice.java: reassigning price in a for-each left [150, 99, 250]; the index loop gave [300, 198, 500].
>
> Run on a Mac, 2026-11-06: Java 25.0.4, Node 24.22.0, TypeScript 7.0.2.

04 TypeScript: for...of, and NaN

for...of gives each value, like Java's for-each. The same <= mistake reads undefined, and 499 + undefined is NaN: no error, a wrong answer.

basket.ts

```
const prices: number[] = [150, 99, 250];
let total = 0;
for (let i = 0; i < prices.length; i++) {
    total += prices[i];
}
console.log(total);    // 499 (NaN with <=)
let sum = 0;
for (const price of prices) {
    sum += price;
}
console.log(sum);      // 499
```

> npx tsc --noEmit was quiet both ways; node printed 499 and 499, then NaN and 499 with <=.
>
> Run on a Mac, 2026-11-06: Java 25.0.4, Node 24.22.0, TypeScript 7.0.2.

Loop over an array

1. Start at 0, the first index.
2. Loop while i < length, never <=.
3. Only need each value? for-each in Java, for...of in TypeScript.
4. An empty array runs the loop zero times.
5. Challenge: start at i = 1 over {100, 200, 300, 400}. 1000 or 900?

Everything else

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The code and the challenge files: complete-modern-java
https://github.com/code-with-sam-dev/complete-modern-java

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